

Problema 4

$$1) f(t) = \sin^2 6t$$

$$\left. \begin{aligned} \sin^2 6t &= 1 - \cos^2 6t \\ \cos(2t) &= 2\cos^2(t) - 1 \end{aligned} \right\} \Rightarrow \sin^2(6t) =$$

$$= 1 - \frac{\cos(12t + 1)}{2} = \frac{1}{2} - \frac{1}{2} \cos(12t)$$

$$\mathcal{L}(f(t)) = \mathcal{L}\left(\frac{1}{2} - \frac{1}{2} \cos(12t)\right)$$

$$= \frac{1}{2} \frac{1}{p} - \frac{1}{2} \frac{p}{p^2 + 12^2} = \frac{1}{2} \left(\frac{1}{p} - \frac{p}{p^2 + 144} \right)$$

$$b) f(t) = \cos^2(3t)$$

$$\cos(2t) = \cos^2(2t) - \sin^2(2t) \Rightarrow \cos^2(t)$$

$$= \cos(2t) + 1$$

$$\mathcal{L}(f(t)) = \mathcal{L}\left(\cos(2t) \frac{1}{2} + \frac{1}{2}\right) = \mathcal{L}\left(\frac{1}{2}\right)$$

$$+ \frac{1}{2} \mathcal{L}(\cos(2t)) = \frac{1}{2} \frac{1}{p} + \frac{1}{2} \frac{p}{p^2 + 2^2} = \frac{1}{2} \left(\frac{1}{p} + \frac{p}{p^2 + 4} \right)$$

$$\frac{p}{p^2 + 4}$$

$$c) f(t) = \sin(3t) \sin(4t) =$$

$$= \frac{1}{2} (\cos(3t-4t) - \cos(3t+4t))$$

$$= \frac{1}{2} (\cos(-t) - \cos(7t)) = \frac{1}{2} \cos t - \frac{1}{2} \cos(7t)$$

$$\mathcal{L}(f(t)) = \frac{1}{2} \mathcal{L}(\cos t) - \frac{1}{2} \mathcal{L}(\cos(7t))$$

$$= \frac{1}{2} \frac{p}{p^2+1} - \frac{1}{2} \frac{p}{p^2+49} = \frac{p}{2} \left(\frac{1}{p^2+1} - \frac{1}{p^2+49} \right)$$

$$d) f(t) = \cos 2t \cos 6t$$

$$f(t) = \frac{1}{2} (\cos(2t-6t) + \cos(2t+6t))$$

$$= \frac{1}{2} \cos(-4t) + \frac{1}{2} \cos 8t = \frac{1}{2} \cos 4t$$

$$+ \frac{1}{2} \cos 8t \Rightarrow \mathcal{L}(f(t)) = \frac{1}{2} \mathcal{L}(\cos 4t)$$

$$+ \frac{1}{2} \mathcal{L}(\cos 8t) = \frac{1}{2} \left(\frac{p}{p^2+16} + \frac{p}{p^2+64} \right)$$

$$= \frac{p}{2} \left(\frac{1}{p^2+16} + \frac{1}{p^2+64} \right)$$

$$1) f(t) = \cos 3t + \sin t = \frac{1}{2} (\sin(3t+4t) - \sin(3t-4t)) = \frac{1}{2} \sin(7t) + \frac{1}{2} \sin(t)$$

$$\begin{aligned} \mathcal{L}(f(t)) &= \frac{1}{2} \mathcal{L}(\sin(7t)) + \frac{1}{2} \mathcal{L}(\sin(t)) \\ &= \frac{1}{2} \frac{7}{p^2+7^2} + \frac{1}{2} \frac{1}{p^2+1} = \frac{1}{2} \left(\frac{7}{p^2+7^2} + \frac{1}{p^2+1} \right) \end{aligned}$$

$$f) f(t) = e^{2t} \sin(3t)$$

$$\mathcal{L}(e^{2t} \sin(3t)) = G(p-2) = \frac{3}{(p-2)^2+9}$$

$$\text{note } Q(t) = \mathcal{L}(g(t)) = \mathcal{L}(\sin(3t)) = \frac{3}{p^2+9}$$

$$g) f(t) = e^{-5t} \cos 3t$$

$$s=5 \Rightarrow f(t) = \frac{p}{p^2+9}$$

$$\mathcal{L}(f(t)) = G(p+5) = \frac{p+5}{(p+5)^2 + 4}$$

$$h) f(t) = e^{-2t} \sinh(4t)$$

$$\cancel{f(t) = e^{-2t} \sinh(4t)}$$

$$\alpha = 2 \Rightarrow f(t) = \sinh(4t)$$

$$G(t) = \mathcal{L}(g(t)) = \frac{4}{p^2 + 16}$$

$$F(t) = \mathcal{L}(f(t)) = G(p+2) =$$

$$= \frac{4}{(p+2)^2 + 16}$$

$$i) f(t) = e^{-t} t^4$$

$$f(t) = e^{-t} t^4$$

$$\alpha = 1, f(t) = t^4$$

$$G(t) = \mathcal{L}(g(t)) = \mathcal{L}(t^4) = \frac{4!}{p^5}$$

$$= \frac{24}{p^5}$$

$$\frac{4!}{p^5}$$

$$f(t) = L(f(t)) = G(p, 0) = \frac{2s}{(p+1)^5}$$

$$j) f(t) = e^{4t} \ln t$$

$$\alpha = -4, g(t) = \ln(2t)$$

$$G(t) = L(g(t)) = L(\ln(2t)) = \frac{p}{p^2 - 4}$$

$$G(t) = L(f(t)) = G(p-4) = \frac{p-4}{(p-4)^2 - 4}$$

$$k) f(t) = e^{3t} t^3$$

$$f(t) = e^{-3t} t^3$$

$$\alpha = 3, f(t) = t^3 \Rightarrow G(t) = L(g(t))$$

$$= L(t^3) = \frac{3!}{p^4} = \frac{6}{p^4} \Rightarrow F(t) = L(f(t))$$

$$= G(p+3) = \frac{6}{(p+3)^4}$$

$$= \frac{6}{(p+3)^4}$$

$$l) f(t) = \sin(5t-10)$$

$$f(t) = \sin(5t)$$

$$f(t) = g(t-2)$$

$$\mathcal{L}(f(t)) = \mathcal{L}(g(t-2)) = e^{-p \cdot 2} \mathcal{L}(\sin(5t))$$

$$= e^{-2p} \frac{5}{p^2 + 25}$$

$$m) f(t) = \cos(4t+8)$$

$$f(t) = \cos(4t+8) \\ = \cos(4t)$$

$$f(t) = g(t+2) = g(t-(-2))$$

$$F(t) = \mathcal{L}(f(t)) = \mathcal{L}[g(t-(-2))]$$

$$= e^{-p \cdot (-2)} \cdot \mathcal{L}(\cos(4t)) = e^{2p} \frac{p}{p^2 + 16}$$

$$n) f(t) = \sinh(3t + 7)$$

~~$$f(t) = \sinh$$~~

$$g(t) = \sinh(3t)$$

$$f(t) = g\left(t + \frac{7}{3}\right) = f\left(t - \left(-\frac{7}{3}\right)\right)$$

$$\begin{aligned} F(s) &= \mathcal{L}\{f(t)\} = \mathcal{L}\left\{g\left(t - \left(-\frac{7}{3}\right)\right)\right\} = \\ &= e^{-p\left(-\frac{7}{3}\right)} \mathcal{L}\{\sinh(3t)\} = e^{p \cdot \frac{7}{3}} \underbrace{\frac{3}{p^2 - 9}} \end{aligned}$$

$$\begin{aligned} o) f(t) &= \cosh(2t - 5) \\ &= \cosh(2t) \end{aligned}$$

$$f(t) = g\left(t - \frac{5}{2}\right)$$

$$\begin{aligned} F(s) &= \mathcal{L}\{f(t)\} = \mathcal{L}\left\{g\left(t - \frac{5}{2}\right)\right\} = e^{-p \cdot \frac{5}{2}} \\ &\cdot \mathcal{L}\{\cosh(2t)\} = e^{-\frac{p \cdot 5}{2}} \underbrace{\frac{p}{p^2 - 4}} \end{aligned}$$

$$p) f(t) = t \sinh(3t)$$

$$f(t) = t \cdot \sinh(3t) = (t)' \cdot \sinh(3t)$$

$$= \sinh(3t) \Rightarrow G(s) = \mathcal{L}(f(t)) = \frac{3}{p^2 - 9}$$

$$F(s) = \mathcal{L}(f(t)) = -\mathcal{L}((1-t)' \cdot \sinh(3t))$$

$$= -\mathcal{L}(\sinh(3t))' = -\left(\frac{3}{p^2 - 9}\right)' = \frac{3 \cdot 2 \cdot p}{(p^2 - 9)^2}$$

$$F(s) = \frac{6p}{(p^2 - 9)^2}$$

$$q) f(t) = t^2 \sin t = (-t^2)' \sin t$$

$$F(s) = \mathcal{L}(f(t)) = \mathcal{L}\{(-t^2)' \sin t\}$$

$$= [\mathcal{L}(\sin t)]'' = \left(\frac{1}{p^2 + 1}\right)'' = \left(\frac{-2p}{(p^2 + 1)^2}\right)'$$

$$= \frac{-2p^2 + 1)^2 + 2p \cdot 2(p^2 + 1) \cdot 2p}{(p^2 + 1)^3}$$

$$= \frac{-2p^2 + 8p^2}{(p^2 + 1)^3} = \frac{6p^2 - 2}{(p^2 + 1)^3}$$

$$4) f(t) = t^3 e^{2t}$$

$$= -(-t)^3 e^{2t}$$

$$F(t) = \mathcal{L}(f(t)) = -\mathcal{L}(t^3 e^{2t}) =$$

$$= (-\mathcal{L}(e^{2t}))^{(3)} = \left(\frac{-1}{p-2} \right)^{(3)}$$

$$= \left(\frac{1}{(p-2)^2} \right)^{(3)} = \left(\frac{-2(p-2) \cdot 1}{(p-2)^3} \right)' = \left(\frac{-2}{(p-2)^2} \right)'$$

$$= \frac{2 \cdot 3(p-2)^{-3}}{(p-2)^4} = \frac{6}{(p-2)^4}$$

$$5) f(x) = \frac{\sinh x}{x}$$

$$F(x) = \mathcal{L}(f(x)) = \mathcal{L}\left(\frac{\sinh(x)}{x}\right) =$$

$$= \int_0^\infty F(q) dq \Rightarrow F(q) = \mathcal{L}(\sinh(x))$$

$$\sim \frac{1}{p^2+1} \Rightarrow F(p) = \int_p^\infty \frac{1}{q^2+1} dq$$

$$= \frac{1}{2} \ln \left(\frac{q-1}{q+1} \right) \Big|_p^\infty = \frac{1}{2} \ln \left(\frac{1-\frac{1}{\infty}}{1+\frac{1}{\infty}} \right)$$

$$= \ln \left(\frac{p-1}{p+1} \right) = \frac{1}{2} \left(-\ln \left(\frac{p+1}{p-1} \right) \right)$$

(2)

$$b) f(t) = \frac{1-\cos t}{t}$$

$$F(p) = \mathcal{L}(f(t)) = \mathcal{L} \left(\frac{1-\cos t}{t} \right)$$

$$= \mathcal{L} \left(\frac{1}{t} \right) - \mathcal{L} \left(\frac{\cos t}{t} \right) = \int_p^\infty \mathcal{L}(1-\cos t) dq$$

$$= \int_p^\infty \left(\frac{1}{q} - \frac{q}{q^2+1} \right) dq = \ln q \Big|_p^\infty$$

$$= \frac{1}{2} \ln(q^2+1) \Big|_p^\infty$$

$$u) f(t) = \frac{\sin t}{t}$$

$$F(p) = \mathcal{L}(f(t)) = \mathcal{L} \left(\frac{\sin t}{t} \right) =$$

$$= \int_p^\infty \mathcal{L}(\sin t) dq = \int_p^\infty \frac{1}{s^2 + 1} dq$$

$$= \arctan(q) \Big|_p^\infty = \frac{\pi}{2} - \arctan p$$

$$(2) a) F(s) = \frac{s+3}{s^2+4} = \frac{s}{s^2+4} + \frac{3}{s^2+4}$$

$$= \frac{s}{s^2+4} + \frac{3}{2} \cdot \frac{2}{s^2+4} = \mathcal{L}(\cos(2t) + \frac{3}{2} \sin(2t))$$

$$\Rightarrow f(t) = \cos(2t) + \frac{3}{2} \sin(2t)$$

$$b) F(s) = \frac{3s-5}{s^2+6s+5}$$

$$s^2+6s+5=0 \Rightarrow \Delta=36-20=16$$

$$s_{1,2} = \frac{-6 \pm 4}{2} \Rightarrow s_1 = -5$$

$$s_2 = -1$$

$$F(s) = \frac{3s-5}{(s+5)(s+1)} = \frac{A}{s+5} + \frac{B}{s+1}$$

$$= \frac{A(s+1) + B(s+5)}{(s+5)(s+1)} \Rightarrow \begin{cases} A+B=3 \\ A+5B=-5 \end{cases}$$

$$-4B=8 \Rightarrow B=-2$$

$$\Rightarrow A=3-B=5$$

$$F(s) = \frac{5}{s+5} - \frac{2}{s+1} = \frac{5}{s+5} - \frac{2}{s+1}$$

$$= \mathcal{L}(5e^{-5t} - 2e^{-t}) \Rightarrow f(t) = 5e^{-5t} - 2e^{-t}$$

$$c) F(s) = \frac{2s+7}{s^2(s-9)}$$

$$F(s) = \frac{2s+7}{s^2(s-9)} = \frac{s(s-9)}{s^2(s-9)} = \frac{s^2-s}{s^2(s-9)} = \frac{s^2(s-9)}{s^2(s-9)} = \frac{s^2(s-9)}{s^2(s-9)}$$

$$= \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s-9}$$

$$= \frac{As^3 - As^2 + Bs^2 - Bs + Cs^3 + C3s^2 + Ds^3 - Ds^2}{s^4 - 8s^3}$$

$$A + B + D = 0$$

$$B + 3C - 3D = 0 \quad | :3 \Rightarrow$$

$$A = 2 \Rightarrow A = -\frac{2}{3}$$

$$-B = 7 \Rightarrow B = -\frac{7}{3}$$

$$A + C + D = 0$$

$$\frac{B}{3} + C - D = 0$$

$$C = -A - \frac{B}{3} =$$

$$= \frac{\frac{2}{3} + \frac{7}{3} \cdot 3}{2} = \frac{6+7}{6 \cdot 3} = \frac{13}{54}$$

$$D = A - C = \frac{2}{3} - \frac{13}{54} = \frac{12-13}{54} = -\frac{1}{54}$$

$$\Rightarrow f(s) = \frac{-2}{s^3} - \frac{7}{s^2} + \frac{13}{s^4(s-3)} - \frac{1}{s^4(s+3)}$$

$$= -\frac{2}{3} \cdot \frac{1}{s} - \frac{7}{3} \cdot \frac{1}{s^2} + \frac{13}{54} \cdot \frac{1}{s-3} - \frac{1}{54} \cdot \frac{1}{s+3}$$

$$= \left(-\frac{2}{3} \cdot 1 - \frac{7}{3} \cdot t + \frac{13}{54} \cdot e^{3t} - \frac{1}{54} \cdot e^{-3t} \right)$$

$$\Rightarrow f(t) = -\frac{2}{3} - \frac{7}{3}t + \frac{13}{54}e^{3t} - \frac{1}{54}e^{-3t}$$

$$d) F(s) = \frac{2s+7}{s^4(s^2+9)}$$

$$= \frac{A}{s} + \frac{B}{s^2} + \frac{Cs+D}{s^2+9}$$

$$= \frac{As^3 + 9As + Bs^2 + 9B + Cs^3 + Ds^2}{s^4(s^2+9)}$$

$$A+C=0 \Rightarrow C = -\frac{2}{9}$$

$$B+D=0 \Rightarrow D = -\frac{7}{9}$$

$$9A=2 \Rightarrow A = \frac{2}{9}$$

$$9B=7 \Rightarrow B = \frac{7}{9}$$

$$\Rightarrow F(s) = \frac{2}{9} \cdot \frac{1}{s} + \frac{7}{9} \cdot \frac{1}{s^2} - \frac{2s+7}{9(s^2+9)}$$

$$= \frac{2}{9} \cdot \frac{1}{s} + \frac{7}{9} \cdot \frac{1}{s^2} - \frac{2}{9} \cdot \frac{s}{s^2+9} - \frac{7}{27} \cdot \frac{3}{s^2+9}$$

$$= \mathcal{L}^{-1} \left[\frac{2}{9} \cdot 1 + \frac{7}{9} \cdot t - \frac{2}{9} \cdot \cos(3t) - \frac{7}{27} \sin(3t) \right]$$

$$2) F(s) = \frac{2s+5}{(s^2-4)(s^2-8s+12)^3}$$

$$= \frac{A}{s-2} + \frac{B}{s+2} + \frac{C}{s-2} + \frac{D}{s-6} \Rightarrow F(s) = -\frac{1}{8} \cdot \frac{1}{s-2} -$$

$$-\frac{9}{16} \cdot \frac{1}{(s-2)^2} + \frac{17}{128} \cdot \frac{1}{s-6} - \frac{1}{128} \cdot \frac{1}{s+2}$$

$$\frac{1}{(p-2)^2} \Rightarrow A(p)=1$$

$$B(p)=(p-2)^2$$

$p_1=p_2=2 \Rightarrow$ rad. dubel

$$\mathcal{L}^{-1} \left(\frac{1}{(p-2)^2} \right) = \frac{1}{(2-1)!} \left[\frac{(p-2)^2}{(p-2)^2} \cdot e^{pe} \right]' \Big|_{p=2}$$

$$= \frac{1}{1!} e^{pe} \cdot e \Big|_{p=2} = e^{2e} \cdot e \Rightarrow f(x) = \frac{1}{8} e^{2x}$$

$$-\frac{9}{16} \cdot e^{2x} \cdot e + \frac{17}{128} \cdot e^{6x} - \frac{1}{128} e^{-2x}$$

$$p) F(s) = \frac{s^2 + 4}{(s-2)(s+4)(s-3)}$$

$$A(p) = s^2 + 4 \Rightarrow \text{grad } A = 2$$

$$B(p) = (s-2)(s+4)(s-3) \Rightarrow \text{grad } B = 3$$

$$B(p) = 0 \quad \begin{cases} s_1 = 2 \\ s_2 = -4 \\ s_3 = 3 \end{cases}$$

$$f(t) = \frac{A_2}{B'(2)} e^{2t} + \frac{A(-4)}{B'(-4)} e^{-4t} + \frac{A_3}{B'(3)} e^{3t}$$

$$B'(s) = (s^3 - s^2 - 15s + 24)' = 3s^2 - 2s - 15$$

$$A(s) = s^2 + 4$$

$$f(t) = \frac{8}{12-4-15} e^{2t} + \frac{16 \times 4}{48+8-15} e^{-4t}$$

$$+ \frac{9+4}{27-6-15} e^{3t} = -\frac{8}{6} e^{2t} + \frac{20}{52} e^{-4t}$$

$$+ \frac{13}{7} e^{3t} = -\frac{4}{3} e^{2t} + \frac{10}{21} e^{-4t} + \frac{13}{7} e^{3t}$$

$$g) f(s) = \frac{2s^2 - 3s + 1}{(s^2 + 1)(s^2 + 4)}$$

$$f(s) = \frac{3s + 7}{3(s^2 + 4)} - \frac{3s + 1}{3(s^2 + 1)} = \frac{s}{s^2 + 4} + \frac{7}{3 \cdot 2}$$

$$= \frac{2}{s^2 + 4} - \frac{s}{s^2 + 1} - \frac{1}{3} \frac{1}{s^2 + 1} = \mathcal{L}(\cos(2t))$$

$$+ \frac{7}{6} \sin(2t) - \cos(t) - \frac{1}{3} \sin(t) \Rightarrow f(t) =$$

$$= \cos(2t) + \frac{7}{6} \sin(2t) - \cos(t) - \frac{1}{3} \sin(t)$$

$$(3) a) x'' + 3x' + 2x = 20e^{3t}, x(0) = 0, x'(0) = 0$$

$$~~x'' + 3x' + 2x = 20e^{3t}~~$$

$$\mathcal{L}(x'' + 3x' + 2x) = \mathcal{L}(20e^{3t})$$

$$\mathcal{L}(x'') + 3\mathcal{L}(x') + 2\mathcal{L}(x) = 20\mathcal{L}(e^{3t}) =$$

$$= 20 \frac{1}{p-3}$$

$$\text{Not } Y(p) = L(x)$$

$$L(x') = p Y(p) - x(0) = p Y(p)$$

$$L(x'') = p^2 Y(p) - \left[p \underbrace{x(0)}_0 + \underbrace{x'(0)}_0 \right] = p^2 Y(p)$$

$$p^2 Y(p) + 3p Y(p) + 2 Y(p) = \frac{20}{p-3}$$

$$Y(p) (p^2 + 3p + 2) = \frac{20}{p-3}$$

$$Y(p) = \frac{20}{(p-3)(p+2)(p+1)} = L(x(e))$$

$x(t) \Rightarrow \text{sol } x.$

$$A(p) = 20$$

$$B(p) = (p-3)(p+2)(p+1)$$

$$B(p) = 0 \Rightarrow \begin{cases} p_1 = 3 \\ p_2 = -2 \\ p_3 = -1 \end{cases}$$

$$x(t) = \frac{A(3)}{B'(3)} e^{3t} + \frac{A(-2)}{B'(-2)} e^{-2t} + \frac{A(-1)}{B'(-1)} e^{-t}$$

$$B'(p) = (p^3 - 7p - 6)' = 3p^2 - 7, A = 20$$

$$x(t) = \frac{20}{27-7} e^{3t} + \frac{20}{12-7} e^{-2t} + \frac{20-5}{5-7} e^{-t}$$

$$x(t) = e^{3t} + 4e^{-2t} - 5e^{-t}$$